

Figure 3.8-8 Process control section of the screen

3. All wire feeders should feed at the base speed of 20 cm/min.

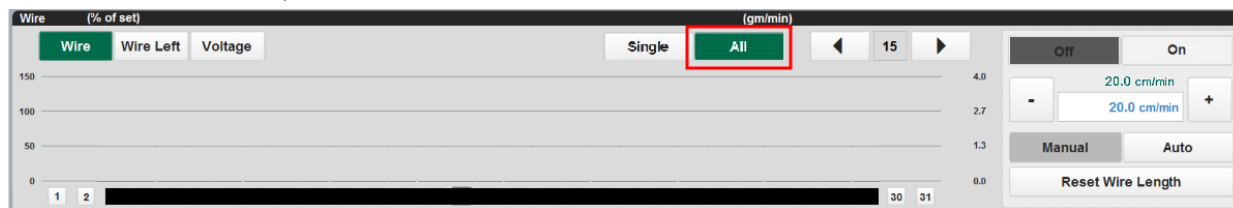
	Note!
	Only the wire feeders selected will be powered up. To drive all the feeders, select the maximum with in the RECIPE HMI screen.

Operation of an Individual Wire Feeder

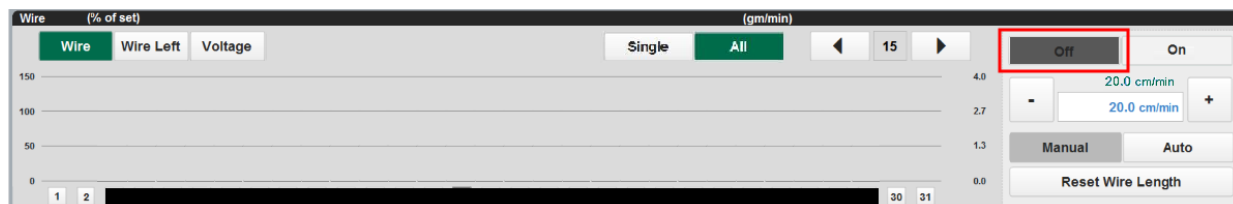
The individual wire feeders can be operated in a similar method, by reducing the individual outputs to zero, except the one being tested.

This is carried out by following these steps:-

1. From the above condition, select ALL button.



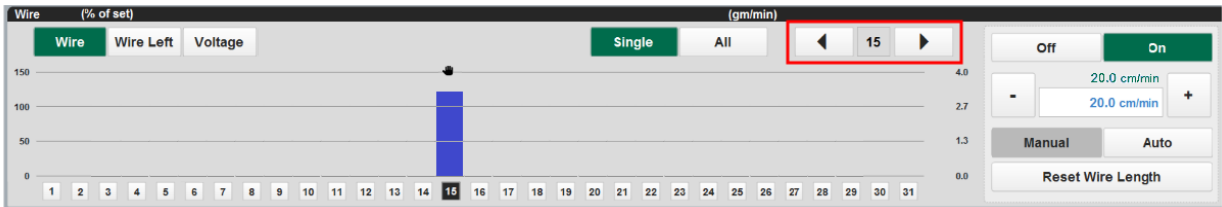
2. Select the OFF button.



3. Select the Single button.



4. Move the cursor to below the channel in question and using the keyboard "arrow" cursors to raise lower the output.



5. The cursor will move to the position and the channel number will change.



6. Select the ON button and the output will be shown at 100%



	Switches the selected channels to 0% output
	Switches the selected channels to 100% output, normally selected
	Decreases the base speed parameter value
	Increases the base speed parameter value
135.0 cm/min	Set point – base speed
0.0 cm/min	Feedback value – base speed
	Allows the user to manually choose a wire feeder channel to select in manual mode of operation while others are in automatic control. Only operational when the chamber is closed and the OD control is in the AUTO mode.
	Places and manual selected channel back into automatic control by the machine. Only operational when the chamber is closed and the OD control is in the AUTO mode.
	Resets the selected channel(9s) to 100% output

Table 3.8-1 Description of wire feeder control buttons.

Selecting Multiple Channels

There is a function designed into the control of the machine to allow selection of all or individual channels.

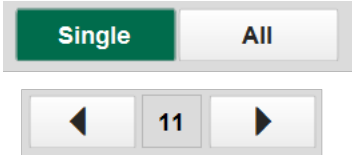


Figure 3.8-9 Channel selection controls

The SINGLE button allows selection of a single channel, and the ALL button selects all. In the case of a SINGLE channel mode, the cursor can be moved using the horizontal arrows shown

Operation Under Normal Production Conditions

Ensure that the RECIPE screen has had the following parameters inputted to allow correct evaporators and wire feeders to be selected.

- Substrate Width
- Nominal Wire Speed

Before pressing the WIRE FEED button, the BOATS ON button must be pressed. Allow approximately four minutes for the boats to heat up before pressing WIRE FEEDERS ON.

The evaporators heat up to a level where they will evaporate the aluminium being fed to them. If auto-wetting is selected, the machine will run the auto-wetting sequence. If auto-wetting is not selected, the wire will be fed at the set speed. To stop the wire, touch the WIRE FEED off button.

The user can modify the wire feed rates using a combination of the "Nominal cm/min Feed Rate" and the wire feed "Output" percentages. However, once the controller is operating in the AUTO mode, it will make corrections to the "Output" percentages.

At the end of the cycle, the auto-slowdown function in the drive system will shut the wire off when the shutter is closed.

Alternatively, the wire can be stopped by touching the WIRE FEEDERS OFF button.

Operation Under New Evaporators Conditions

If the wire does not melt and evaporate on the evaporators, the power level to them all can be increased. Alternatively the wire rate can be turned down.

If one evaporator is causing problems, remove the wire feed to the faulty evaporator and leave the others running. Heat up the faulty evaporator and try wetting again at a lower speed. This will be tolerated by the other evaporators.

3.9 Shutter System Operation - Evaporation Source

An aluminium water-cooled shutter is located above the source enclosure, when the machine is closed. It protects the substrate from the radiant heat generated by the evaporators whilst they are being heated subsequent to metallizing the substrate.

The shutter is moved by the action of a set of air operated linear cylinders being driven as the line speed approaches its set level. On deceleration, the shutter closes at a similar rate.

The shutter is located below the coating shield with the cylinders built into the assembly.

If the shutter is opened at too low a web speed the substrate can be damaged, a minimum speed is set at 120m/min (non-adjustable). The machine operator has the ability to select an open/closing speed above this value, as a percentage of the selected line speed as a percentage. .

To move the winding cart into the chamber, the shutter has to be in its closed position. Interlocking proximity sensors are supplied for this purpose

	Caution ! The shutter must be in the closed position for the winding cart to be moved towards the chamber.
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Figure 3.9-1 Shutter lowered and in the closed position

The movement of the shutter can be isolated by rotating the cart isolation (G-Stop) switch to the OFF position.

Opening of the Shutter (Moving Shield)

The shutter can be manually or automatically opened and closed.

Manual Operation

This shutter control allows it to be operated to fully closed or fully open position.

The operational mode selection needs to be set for MANUAL mode. [See chapter 1.1 Hawkeye Reporting System on page 1](#)

The shutter can be closed at any time by pressing the CLOSE button.

Automatic Operation

The automatic mode of operation can be selected in the CONFIGURATION MODES 2 HMI screen. It will allow the shutter to be opened automatically as the line speed increases. [See chapter 1.1 Hawkeye Reporting System on page 1](#)

Once the web is running the shutter will be opened automatically as the line speed increases.

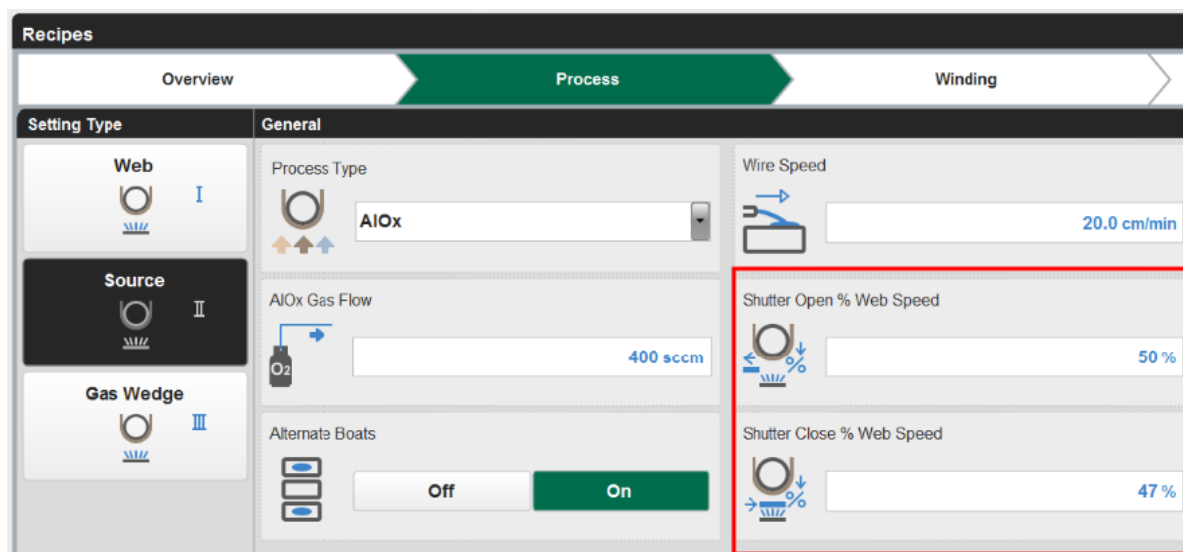


Figure 3.9-2 RECIPE HMI screen

The speed at which the shutter will open is set as a percentage of the line speed from the roll SETUP HMI screen.

The shutter open speed can be set as a percentage of the set line speed. The open speed must be higher than the close speed.

When the actual speed reaches 90% of the set line speed in this example, it will start to open the shutter.

	Note!
	If the shutter open speed is set lower than the minimum speed of 120 m/min, it will not start to open. If the shutter is open at a low speed then the heat from the evaporators will damage the web, by heat and condensed aluminium.

Automatic Closing of the Shutter

As the winding drive system decelerates at the end of the coating cycle, the shutter has to be closed, as described below.

- The shutter will close when the Auto slowdown sequence is started by the drive system. if the auto operation mode has been selected.
- The shutter can be closed mid-coating cycle by touching the SHUTTER CLOSE button. Pressing the close button will take the control out of the automatic mode.

Opening of Shutter for Access

The shutter can be moved to allow maintenance and adjustment to the coating width shields, for example, or for cleaning.

With the coating shield in the vertical position, use the lever switch below the screen to move the shutter to the open position.



Figure 3.9-3 Shutter in the open position

	DANGER !
	Keep clear of the shutter/shield when being operated Lock out the cart isolation (G-Stop) switch to the OFF position, before working in the area.

3.10 Evaporation Source - Viewing

A water-cooled aluminium flap is located on the operator side of the source enclosure. Its purpose is stop unnecessary coating of components outside the source enclosure under normal process conditions.

The second part of the system is a rotating tube called a strobe tube. The purpose of the strobe tube is to allow the operator to view the evaporators without allowing aluminium to coat the viewing window completely.

To view the aluminium wire being fed onto the evaporators, press the View Source button below the HMI screen. This will start the strobe tube rotating and then the flap will open allowing viewing, after a short time delay.

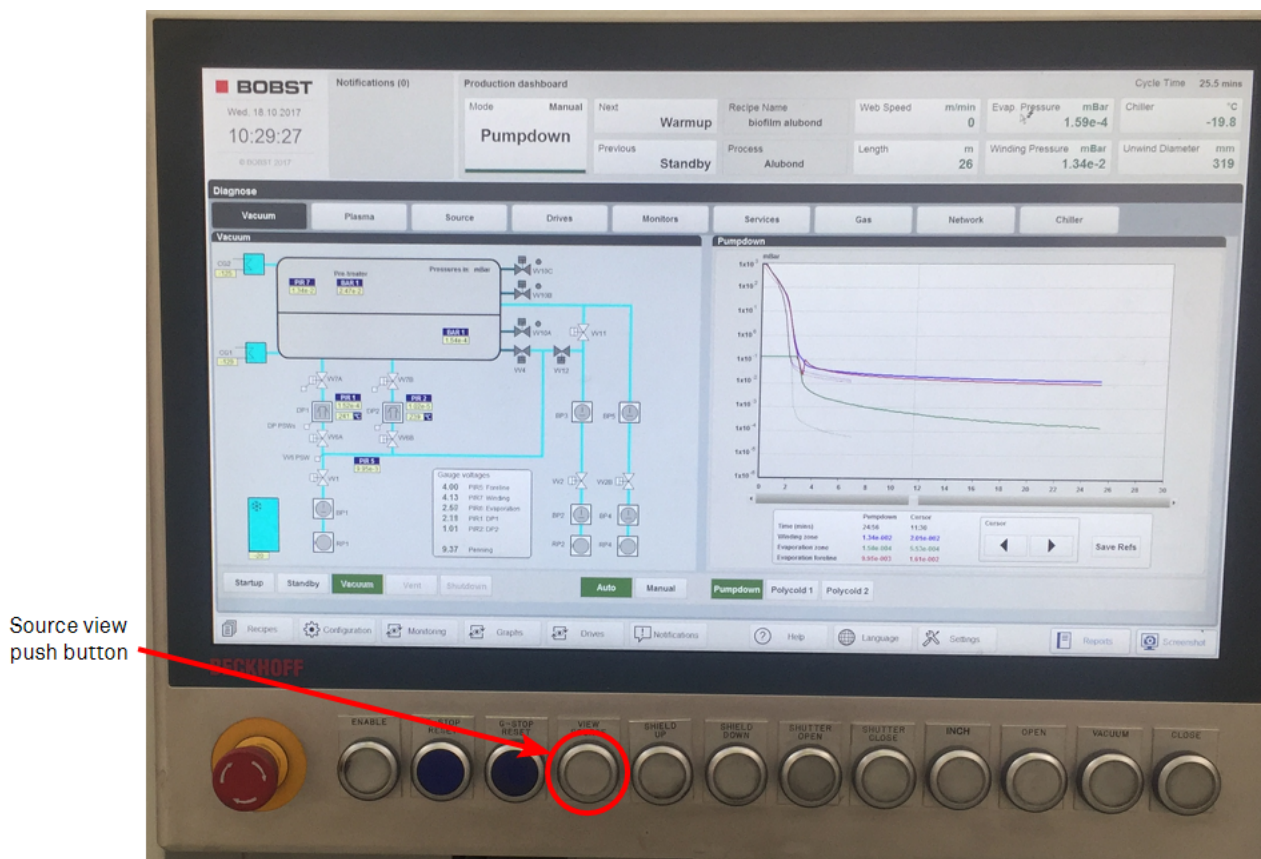


Figure 3.10-1 Shutter control pushbuttons on PROCESS HMI screen

The viewing flap is rotated by the action of a pneumatic actuator located on the outside of the chamber



Figure 3.10-2 Viewing flap movement actuator

	Caution !
	The flap must be in the vertical closed position for the winding cart to be moved towards the chamber. Lock out the cart isolation (G-Stop) switch to the OFF position, before working in the area.

The flap's pneumatic control valves are designed to allow safe working upon the flap and evaporation source enclosure or evaporators when the winding cart is retracted from the chamber, and the cart isolation switch has been rotated to the OFF position.

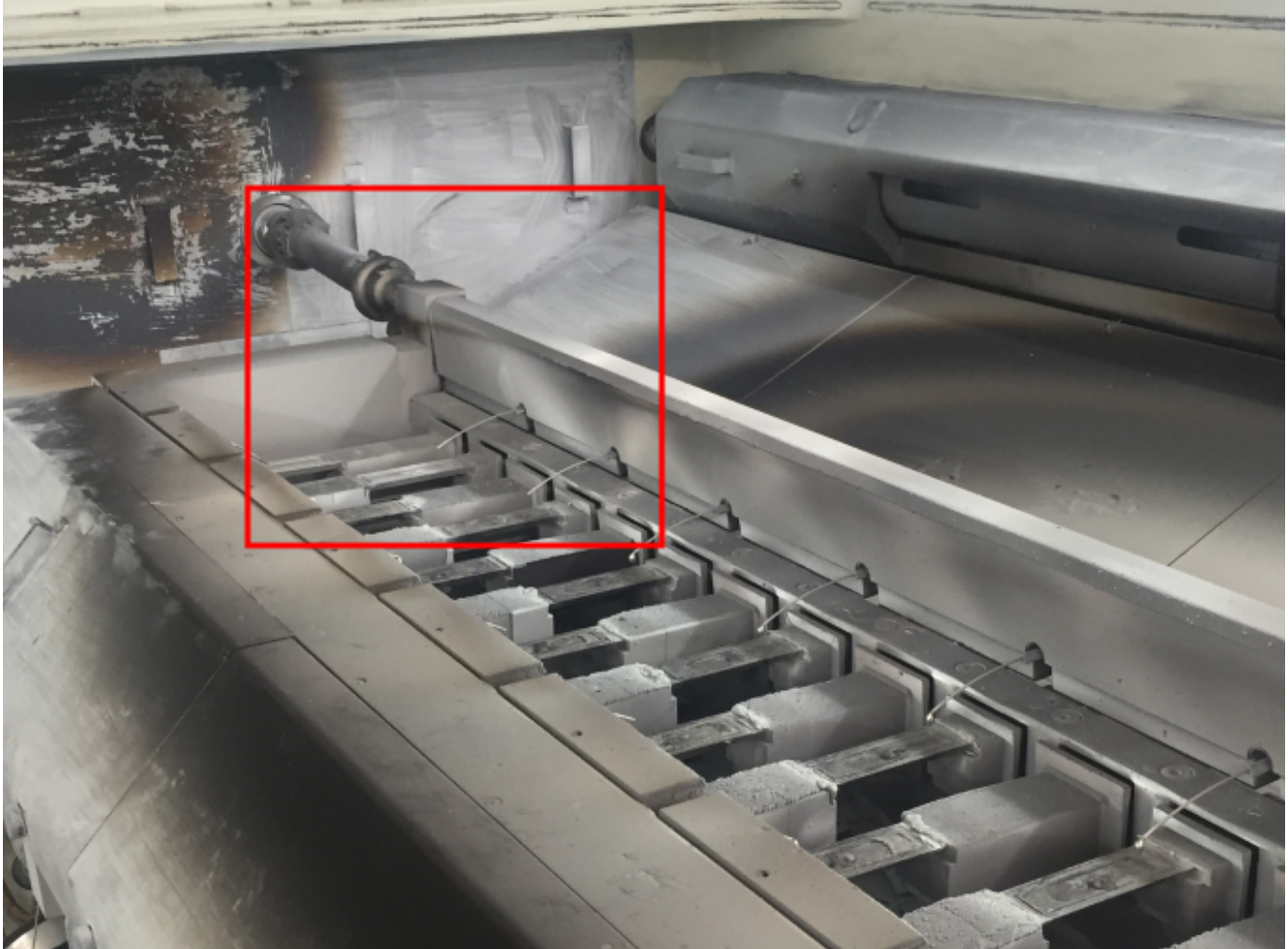


Figure 3.10-3 Evaporation area with the winding cart retracted and the viewing flap on the right-hand side.

User Operation

This strobe control allows it to be operated at any time, once the chamber is closed and power has been applied to the evaporators (Boats Power On button).

Stopping Viewing and Closing of the Flap

A countdown timer will automatically stop the strobe tube rotation and close the viewing flap after approximately 4 minutes.

Alternately, the user can also stop it by again touching the view source button.



Figure 3.10-4 Strobe tube drive motor and coupling

Removing the Strobe Cover

To remove the cover for maintenance/ cleaning purposes.



Warning!

Ensure that the Movement isolator is locked out before attempting to work on this equipment.



Warning!

If there is a requirement to rotate the equipment, be aware that all precautions are taken to ensure that no parts of your body can be trapped.

1. Lockout the “traverse lockout isolator” to ensure that the motor cannot be enabled.



Figure 3.10-5 The complete viewing window strobe assembly

2. Remove collection shields between the evaporation source and the chamber wall to allow access

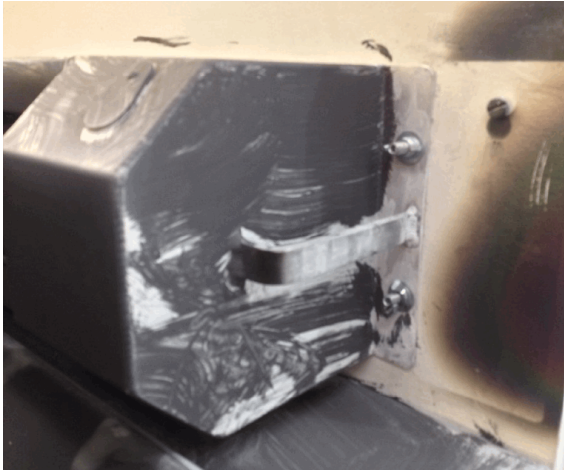


Figure 3.10-6 Fasteners that locate the cover

3. The internal guard has to be removed, by removing the locating wing-nuts.



Figure 3.10-7 Example of a technician removing the fasteners

4. Lift the strobe cover over the strobe and away from the assembly



Figure 3.10-8 Example of a strobe with cover removed.

Make sure that the cable to the motor passes under and through the hole provided in the strobe cover when re-assembling.

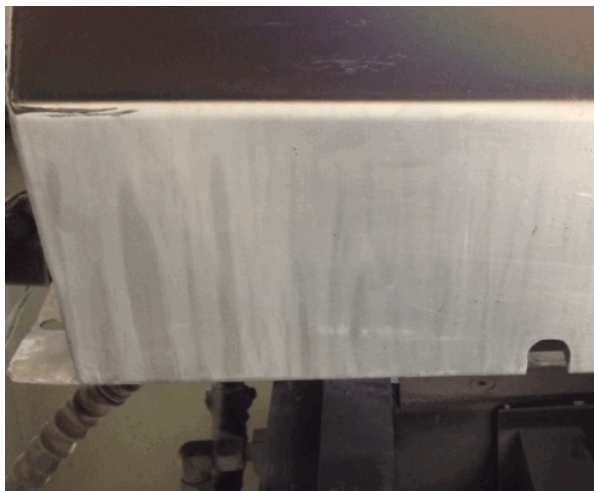


Figure 3.10-9 Hole in strobe cover.

How to remove and clean the strobe viewing glass windows

The strobe viewing glass can be removed for cleaning when the strobe cylinder has been removed, [See chapter Removing the Strobe Cover on page 117 for details.](#)



Figure 3.10-10 Strobe viewing glass window

The glass can be cleaned in two ways:-

- Mechanical, using a sharp blade, as a scraper. Do not damage the glass, as it will become more difficult to clean later.
- Chemical: - Using a weak caustic soda solution, commonly used to clean blocked drainage systems. Safety Guidelines <http://www.dow.com/productsafety/finder/caustic.htm>

	Warning !
	Ensure that care is taken when using either cleaning method and follow the chemical supplier's instructions/guidance. Ensure that the Movement isolator is locked out before attempting to work on this equipment.

To removes the glass:

1. Open the locking clips on the protective housing



Figure 3.10-11 Example of a technician opening the locking clips.

2. Open the protective housing

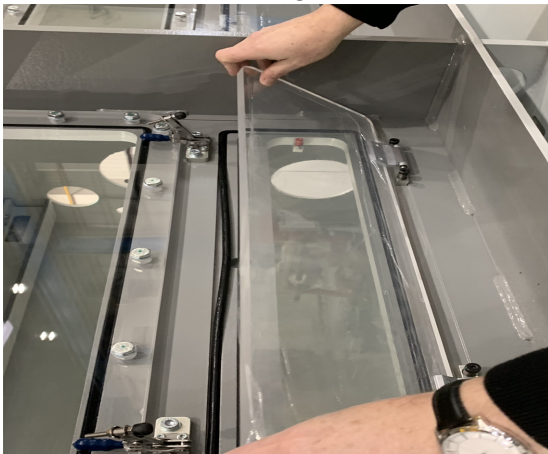


Figure 3.10-12 Example of a technician opening the protective housing.

3. Carefully remove the glass

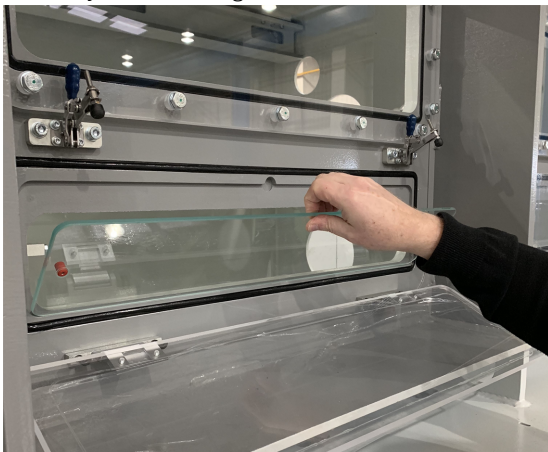


Figure 3.10-13 Example of a technician removing the glass

Use the magnetic pins on the viewing glass to position the viewing glass when re-assembling the window.



Figure 3.10-14 Example of the magnetic locating pins on the viewing glass

3.11 Use of Release Agent

The collection efficiency of the evaporated aluminium onto the passing substrate is approximately 70%. The other 30 % will be evaporated and collected on the shielding around the boats.

In order to prevent evaporated aluminium adhering permanently to the source box, surrounding collection shields and shutter, the surfaces must be painted with a release agent /paint

The recommended release agents are Boron Nitride paint or Aquadag Graphite Paint diluted with water.

Careful and adequate application of the paint will be repaid by easy cleaning of the machine.



Figure 3.11-1 Example of a technician applying release paint

In places such as the underside of the shutter, where some adhesion of the aluminium is required, the paint should be applied more sparingly than on the inside of the source box.



Figure 3.11-2 Shutter with release paint applied

The specification of the release agents can be found in the consumable section of this manual.



Caution !

Do not use hammers or chisels to remove aluminium from the machine's collection surfaces.

Avoid painting the electrical contact surfaces and any evaporators that may be in the source box.



Figure 3.11-3 Evaporation source area painted with release agent



Note !

Places such as the underside of the shutter, where some adhesion of the aluminium is required, the paint should be applied more sparingly than on the inside of the source box.

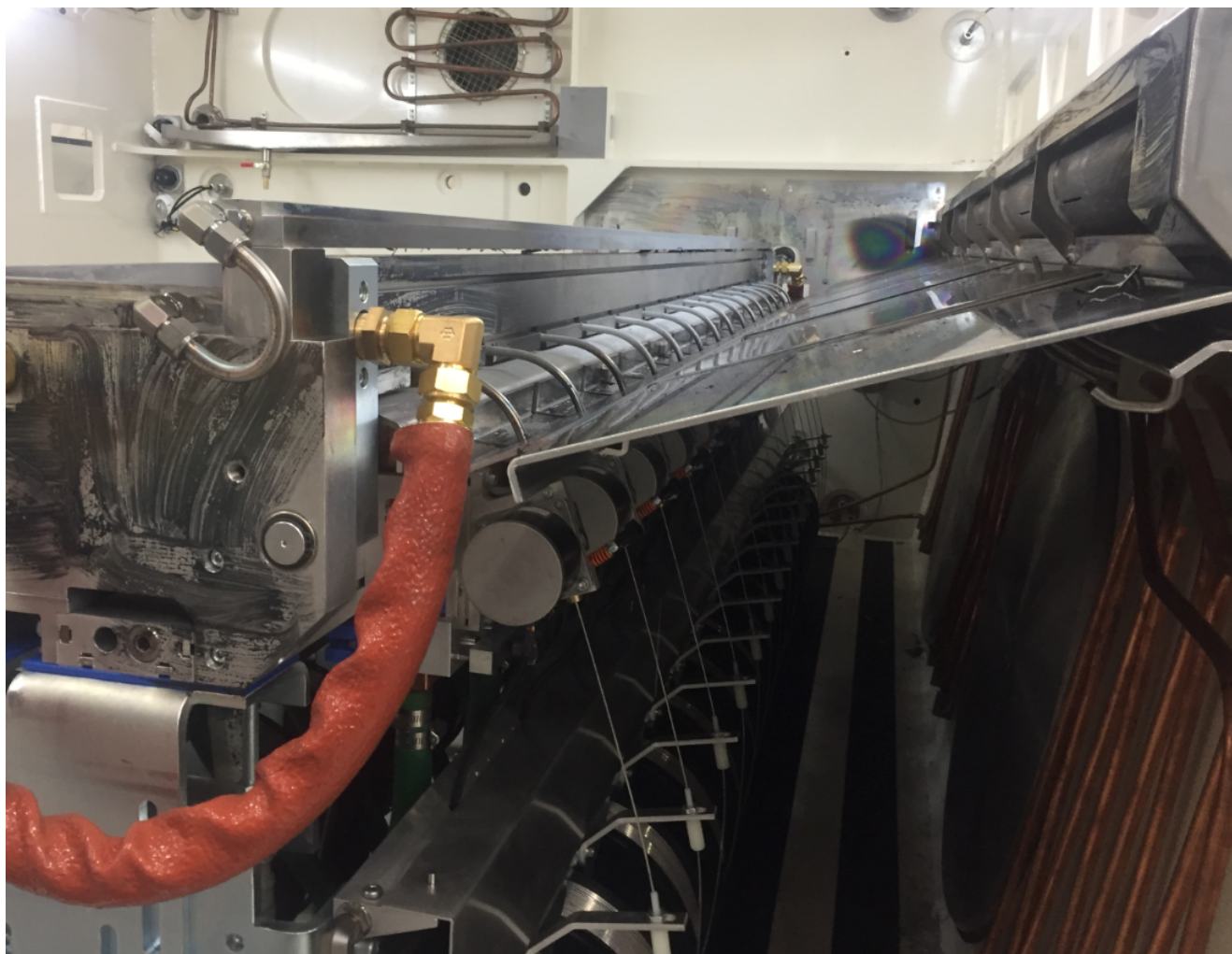


Figure 3.11-4 Removable shielding between the source and the chamber wall after being painted with release agent

3.12 Visual Inspection of Coating Source Area before Closing the Chamber

The evaporators and immediate area must be checked before closing the chamber.

The coating shield, viewing flap and shutter must be also checked and cleaned for loose aluminium coatings.

This includes-

- Checking that there is enough aluminium wire left on the wire spools.
- Cleaning the evaporator surfaces
- Checking the wire feeder tubes for no potential blockages
- Checking that no potential shorts can take place between the wire feeders/tubes and the surrounding steelwork.
- Checking that no loose debris is hanging of the shutter
- Checking that there is no loose aluminium debris that can cause shorts to the evaporators
- Check that all the collection shields are in place.

3.13 Checking Aluminium Wire Spools for Low Wire Levels

The aluminium wire is usually supplied by a commercial supplier on plastic spools.

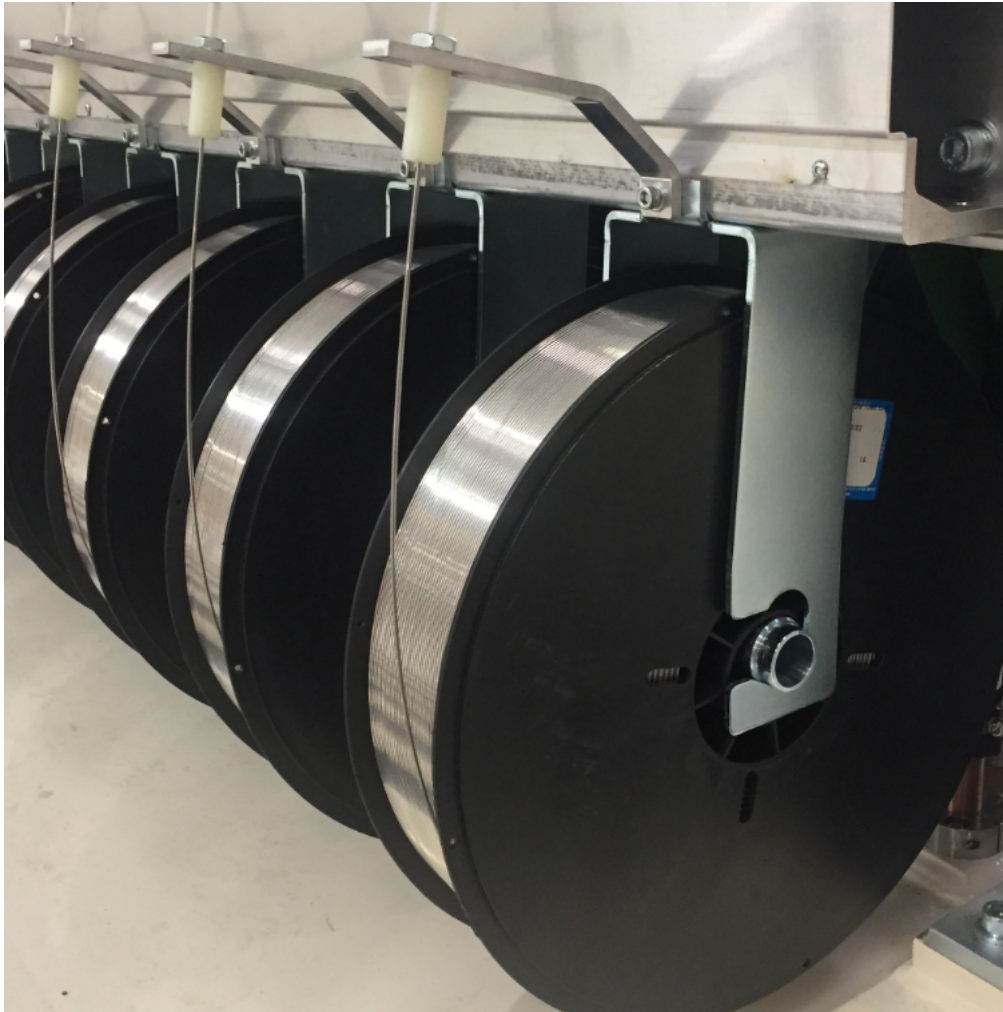


Figure 3.13-1 Example of a full wire spool

The amount of wire is normally measured in weight. This is normally between 8.5 and 14.6 KG.

To enable the user to visually see how much wire is still on a spool, a slot has been designed into the plastic spools.



Figure 3.13-2 Spool side opening used to check how much wire is remaining

On a standard spool, as the wire initially gets to point where the slot is no longer full, there is approximately enough wire for one process cycle (<40 minutes) left.